“PhysioCare”

[Protocol: Exercise for Upper Limb Lymphedema for Breast Cancer Treatment]

Dr Maria Matsangidou.
Research centre on Interactive media,
Smart systems and Emerging Technologies (RISE),
Tryfon Building, 1 Constantinou Palaiologou
Eleftheria Square, 1011, Nicosia, Cyprus.

E-mail: m.matsangidou@rise.org.cy or matsangidou.m@gmail.com
Phone: +357 99524396

1. General information

1. Ideally, we would like the patient to perform the exercises 3 times per day (morning, lunch and noon time), but we can tolerate this to once a day.

    - ***System Takeout:*** The system should remind the patient to perform the exercise tasks at a pre-selected time in the morning, lunch and noon time. To do so use a positive quote: “Let’s train together”, “Ελα να γυμνασθούμε μαζί”, “The workout you do today, determines who you will be tomorrow”, “Χτίσε σήμερα ένα υγιές αύριο”.

      Since not all patient will be happy to perform the training 3 times per day, the system should allow them to dismiss the reminders.

2. Each training session should last approximately 10 minutes and should combine deep breathing neck, shoulder, and arm exercises.

3. The exercises shouldn't be painful. We aim for the patient to sense a stretching feeling during the exercises but not to burden the muscles.

    - ***System Takeout:***

        i. The system should automatically record weather the exercise was **(a)** completed from the user; **(b)** whether difficulties were encountered by the user (e.g., reaction time, completion time, difficulties in approaching the target, etc.);

        ii. The system should record **(c)** via a subjective report given by the user the level of pain felt by the user.

    How to:

Visual Analogue Scale (VAS) (please see below) will presented to the user before (pre-) and after (post-) exercise.

The system will ask the user to record its choice by looking for a few seconds on the relevant emoji.

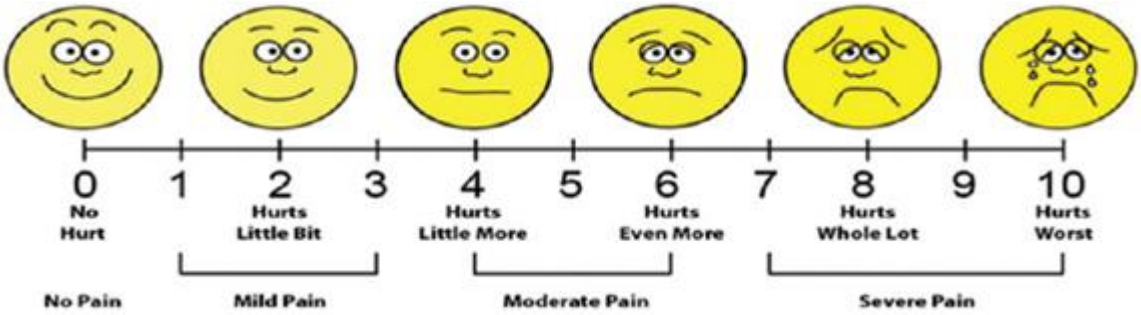
Pre- Evaluation

Please rate the level of pain you are feeling now?

– To rate please look at the emoji of your choice until a tick appear.

Πόσο πόνο αισθάνεσαι τώρα;

– Για να επιλέξεις τον πόνο που αισθάνεσαι κοίταξε για μερικά δευτερόλεπτα το πρόσωπο που σε εκφράζει. Όταν η επιλογή σου καταγραφεί θα εμφανιστεί ένα τικ στην οθόνη.



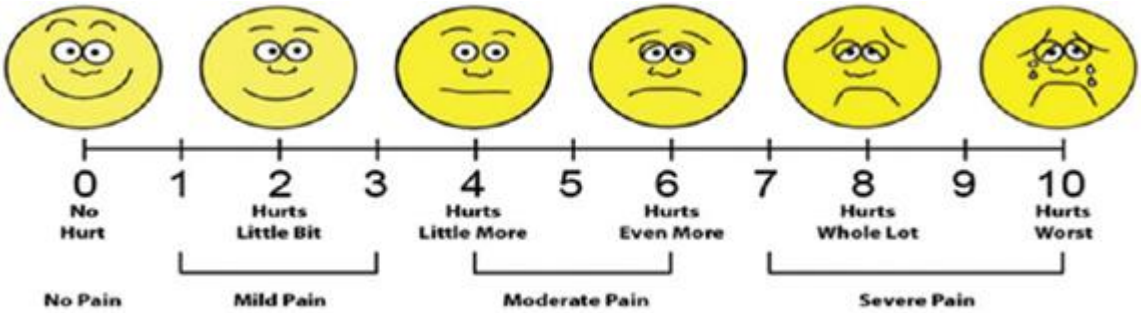
Post- Evaluation

Please rate the level of pain you were feeling while you were doing the exercise task.

– To rate please look at the right emoji until a tick appear.

Πόσο πόνο αισθανόσουνα κατά τη διάρκεια της άσκησης;

– Για να επιλέξεις τον πόνο που αισθάνεσαι κοίταξε για μερικά δευτερόλεπτα το πρόσωπο που σε εκφράζει. Όταν η επιλογή σου καταγραφεί θα εμφανιστεί ένα τικ στην οθόνη.



- ***System Design:*** Use a circle to indicate the user's viewpoint/ eye-gaze. Once the user stays still looking at the preferable emoji for about 5 seconds use a circle loading animation to indicate the choice →

4. Each exercise should repeat at least for 5 times. The preferable is to repeat each exercise for 10 times.

    - ***System Takeout:*** For future reference the system should personalise the repetitions based on the medical record of the patient. At the moment, we will have a pre-set limit which will be 10 repetitions for each exercise task. However, if the user performs only some of the repetitions and then quit the exercise (stay still for 30 seconds) or fail to raise the arm to the exercise level the system should automatically continue to the following exercise.

    - The system should also increase the task difficulty gradually, so after a month the system should offer to the patient 2 sets of 10.

5. During most of the exercises it is suggested for the patient to open and close the fist of the working out hand to the top and the bottom of the exercise. In the exercise it is performed without openings and closings the fist the patient needs to perform ball squeezes, to improve pumps compressions.

    - ***System Takeout:*** Unfortunately, we are not able at the moment to check whether the patients do the openings and closings of the first. This is a limitation set by our equipment.

    - However, we need to add an instruction to remind the patient to perform pumps compressions with the ball once the training is over.

2. Main Exercises

Category: Deep Abdominal Breathing

1. Each session should *start and ends* with deep abdominal breathing. This is because deep abdominal breathing exercise can help the flow of the lymph fluid through the body, and it encourages the lymph to flow back into the blood system. Therefore, deep abdominal breathing exercise is an important step for the treatment and prevention of all lymphoedema types.

    - ***Exercise Steps:***

        - Relax your shoulders and upper chest.
        - Rest one of your hands below your ribs.
        - Take a slow, deep, comfortable breath in through your nose, feeling your hand rise as your tummy (abdomen) rises.
        - Then slowly breathe out through your mouth, so your abdomen is flat again.
        - Repeat for 5 to 10 times.

- ***General Info - System Takeout:*** Enhance the system with calming scenes and sounds (e.g., you can place your user in a calming green environment, next to a peaceful - gentle riverside, or to a shore of a calm sea. You may enhance the scenery with bird signing, or soft instrumental music. Finally, you might add an instructor voice indicating the retheme of the inhale and exhale).

- ***Check out these 360 videos for ideas:***

    i. https://www.youtube.com/watch?v=bW9VYhytk-c

    ii. https://www.youtube.com/watch?v=7AkbUfZjS5k

    iii. https://www.youtube.com/watch?v=MgJITGvVfR0

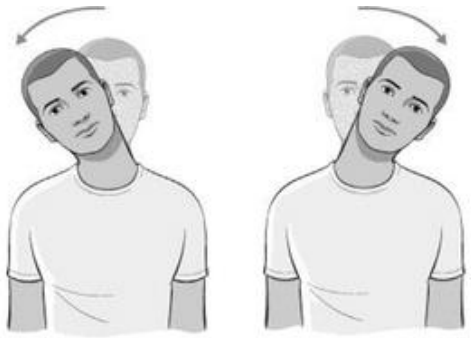
- ***System Takeout:*** Use two points (colliders) to indicate the starting and ending point of each exercise. Once the user touches the first point, the point should stay still for 3 seconds, then decapper or move to the direction the arm should move to indicate the movement and then stay still at the ending point. Once the user brings the arm to the ending point, the point should stay still for 3 seconds and then decapper or move to the direction the arm should move.
  Be consistent to all exercise tasks and repeat this process for ten repetitive times.

Category: Neck exercises

2. ***Exercise: Neck Tilting***

    - ***Exercise Steps:***

      Keep your shoulders still while doing this exercise:
      - Tilt your head towards one shoulder.
      - Hold for 3 seconds.
      - Return your head to the centre.
      - Hold for 3 seconds.
      - Repeat on the other side.
      - Repeat for 5 to 10 times.

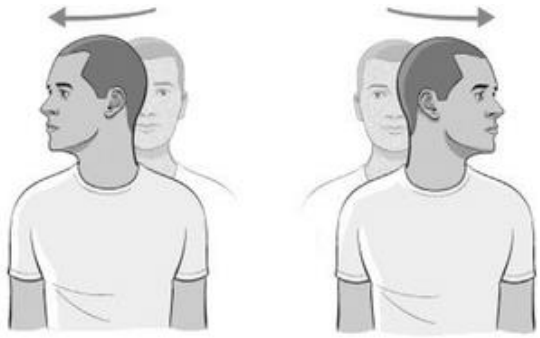


2. ***Exercise: Neck Turning***

    - ***Exercise Steps:***

      Keep your body still while doing this exercise:
      - Turn your head to look over your shoulder as far as you can.
      - Hold for 3 seconds.
      - Return to face forward.

- Hold for 3 seconds.
- Repeat over the other shoulder.
- Repeat for 5 to 10 times.



3. ***Exercise: Chin to Chest***
   - ***Exercise Steps:***

     Keep your body still while doing this exercise:
     - Bend your neck down as far as you comfortably can so your chin is on your chest.
     - Hold for 3 seconds.
     - Bring your head back up.
     - Hold for 3 seconds.
     - Repeat for 5 to 10 times.

Category: Shoulder exercises

4. ***Exercise: Shoulder up and down***

   - ***Exercise Steps:***

     Keep your body still while doing this exercise:

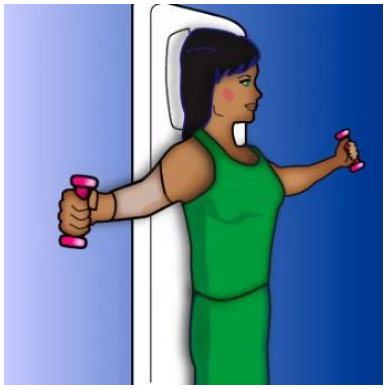
     - Relax your shoulders.
     - Bring them up to your ears.
     - Hold for 3 seconds.
     - Drop them back down.
     - Hold for 3 seconds.
     - Repeat for 5 to 10 times.

5. ***Exercise: Shoulder Adduction***

   - ***Exercise Steps:***

     Keep your back and neck straight and your shoulders relaxed while doing this exercise:

     - Extend your arm away from your body, to your side.
     - Without bending your elbow, slowly bring your arm straight in front of you.
     - Hold for 3 seconds.
     - Then slowly lower your hand back to the first position (straight to your side).
     - Repeat for 5 to 10 times.

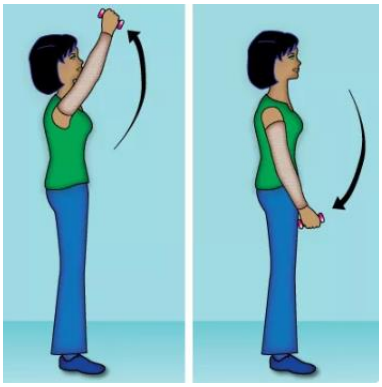


6. ***Exercise: Shoulder Flexion***

   - ***Exercise Steps:***

     Keep your back and neck straight and your shoulders relaxed while doing this exercise:

     - Slowly raise your arm, using a gentle controlled motion.
     - When your arm is almost directly overhead, pause and hold this position for 3 seconds.
     - Now slowly lower your arm—don't swing but use control—until your arm is back beside your body.
     - Rest for 3 seconds.
     - Repeat for 5 to 10 times.



Category: Arm exercises

7. ***Exercise: Arm Combined movements***

   - ***Exercise Steps:***

     Keep your body still while doing this exercise:

     - With your hands touch your thighs.
     - Then touch your shoulders.
     - Then raise your hand up to the sky.
     - Then touch your shoulders again.
     - Then touch your thighs again.
     - Repeat for 5 to 10 times.

8. ***Exercise: Sideways arm raise***

   - ***Exercise Steps:***

     Keep your body still while doing this exercise:

     - Rest your arm by your side. Have your palm to face forward.
     - Then fully raise your arm to the sky.
     - Keep for 3 seconds.
     - Lower your arm to your side.
     - Repeat for 5 to 10 Times.

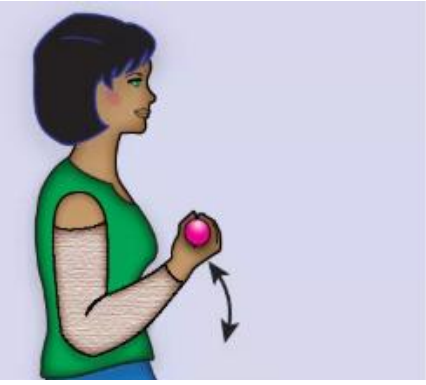
Category: Elbow exercises

9. Exercise: Elbow Flexion - Bicep

- ***Exercise Steps:***

  Keep your back and neck straight and your shoulders relaxed while doing this exercise:

  - Rest your hand on your lap.
  - Then slowly bend your elbow and lift your hand towards your chest.
  - Hold for 3 seconds.
  - Then slowly lower your hands back down to your lap.
  - Repeat for 5 to 10 times.

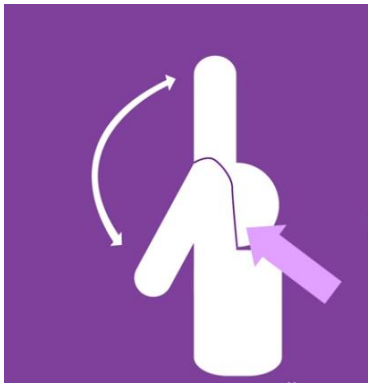


10. Exercise: Elbow Flexion - Triceps

- ***Exercise Steps:***

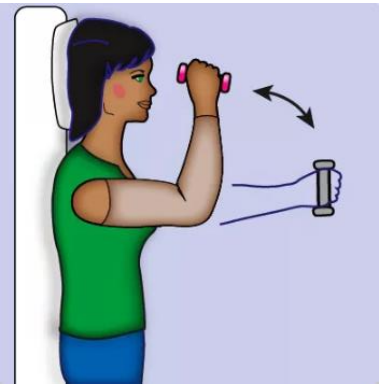
  Keep your back and neck straight and your shoulders relaxed while doing this exercise:

  - Bring your arm up (facing the sky).
  - Then bend your elbows right behind you. – Do not let your arm flop down.
  - Then hold your hand up to that position for about 3 seconds.
  - Then slowly raise your hand back up.
  - Repeat for 5 to 10 times.



11. Exercise: Elbow Extension

- ***Exercise Steps:***

  Keep your back and neck straight and your shoulders relaxed while doing this exercise:

  - Raise your hand straight in front of you.
  - Keep your shoulder still in front of you.
  - Then slowly bend your elbow and lower your hand towards your forehead level.
  - When your elbow is bent at a 90-degree angle, stop moving and hold the position for about 3 seconds.
  - Then slowly lower your hands back to the first position (straight in front of you).
  - Repeat for 5 to 10 times.